Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/05/2004
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Pool Boiling Heat Transfer Mechanisms in Microgravity and Partial Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	12/22/2003	**End Date:**	10/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note: extended 4 more years		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA26A		
Performance Goal No.:			
Performance Goal Text:			
	The objective of the proposed research is to determine the effects of microgravity on nucleate boiling heat transfer. An understanding of microgravity effects on boiling mechanisms is critical to the proper design of phase change heat removal equipment for use in space-based applications. Experiments to date have shown that stable, subcooled boiling on flat plates in microgravity environments is possible. It is presently thought that coalescence and condensation of bubbles away from the heater cause oscillatory motions within the surrounding liquid, thereby providing a continual supply of liquid from the bulk of the pool to the base of the bubble just above the heater. Almost all research pertaining to boiling in microgravity environments, however, has thus far been either of a qualitative nature (photographic studies) with some average wall heat flux/wall temperature measurements, analytical work, or numerical simulations. In the studies where the heat transfer coefficients were measured, the heated surface was always comparable to or much larger than the bubble sizes, so only average heat transfer rates over the entire heated surface were obtained. Very little		

Task Description:	experimental data is available regarding the local heat transfer rates under and around the bubbles as they grow and depart from the surface. Such information, especially when correlated with visual studies, can provide much needed information regarding the relevant wall heat transfer mechanisms during the bubble departure cycle by pinpointing when and where in the cycle large amounts of heat are removed. Microgravity effects on critical heat flux levels (the maximum heat flux level that can be attained without a catastrophic increase in heater temperature) is another very important area that must be addressed if boiling is to be used reliably as a heat removal mechanism. Although critical heat flux is known to decrease with decreasing gravity, very little quantitative information is available due to the use of constant wall heat flux heaters. A four-year project with two primary objectives is proposed: 1) to determine the relative contributions of various heat transfer mechanisms to the overall heat flux in subcooled, nucleate pool boiling on a flat plate in microgravity environments, and 2) to obtain quantitative data on the effects of microgravity on critical heat flux levels. The research will results in better understanding of the mechanisms of boiling heat transfer, and should enable the development of more efficient two-phase heat exchangers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research has shown that transient conduction is the dominant heat transfer mechanism in boiling of refrigerants-like fluids.
Task Progress:	Data taken previously on the KC-135 aircraft using subsets of heaters on a 7.0 mm x 7.0 mm heater array was analyzed. Further data was obtained at various subcoolings. Low-g boiling behavior was governed by the dynamics of the primary bubble. For both square and rectangular heaters, CHF appeared to be a result of the competition between increasing heat transfer associated with the satellite bubbles and the decrease in heat transfer due to growth of the dry area under the primary bubble as the wall superheat increases. At low subcooling on rectangular heaters, surface tension acted to pull the bubble into a spherical shape, allowing liquid to rewet the surface. At high subcooling and high superheat, thermocapillary convection caused the large bubbles that formed on the surface to shrink by increasing the condensation on the bubble cap, resulting in more wetted area. The presence of thermocapillary convection at higher subcooolings may be due to FC-72 being a mixture of various components. Heat transfer under nucleating bubbles in earth gravity were directly measured using a microheater array with 100 _m resolution with low and high subcooled bulk liquid along with images from below and from the side. The array was held at constant temperature. The individual bubble departure diameter and energy transfer were larger with low subcooling but the departure frequency increased at high subcooling, resulting in higher overall heat transfer. The bubble growth for both subcoolings was primarily due to energy transfer from the superheated liquid layer–relatively little was due to wall heat transfer during the bubble growth process. Oscillating bubbles and sliding bubbles were also observed in highly subcooled boiling. Transient conduction and/or microconvection was the dominant heat transfer mechanism in the above cases. A transient conduction model was developed and found to compare well with the experimental data. Time and space resolved temperature and heat flux distributions under nucleating bubbles on a constant heat flux surface were obtained using a 10x10 microheater array with 100 µm resolution along with high-speed images in earth gravity. A numerical simulation was used to compute the substrate conduction, which was then subtracted from the heater power to obtain the wall-to-liquid heat transfer. The data indicated that most of the energy required for bubble growth came from the superheated layer around the bubble. Microlayer evaporation and contact line heat transfer could account for not more than 23% of the total heat transferred from the surface. The dominant heat transfer mechanism was transient conduction into the liquid during bubble departure. Bubble coalescence was not observed to transfer significant amount of heat.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
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Papers from Meeting Proceedings	Henry, C.D., Kim, J., Chamberlain, B., and Hartman, T.G. "Heater size and heater aspect ratio effects on subcooled pool boiling heat transfer in low-g" 3rd International Symposium on Two-Phase Flow Modeling and Experimentation, Pisa Sep-2004